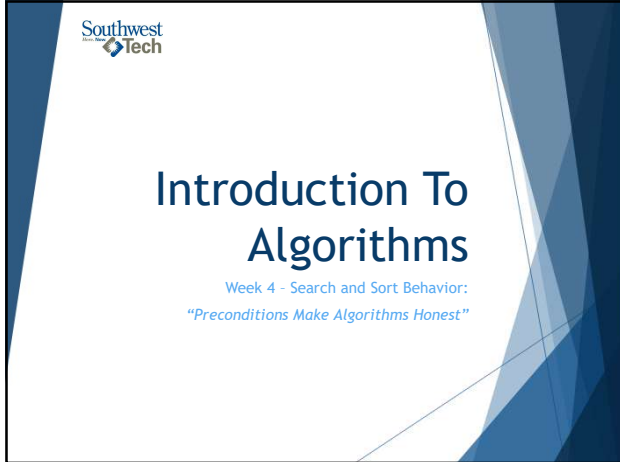
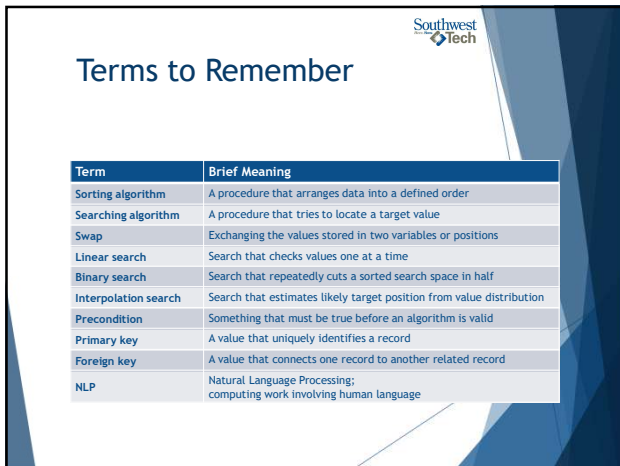




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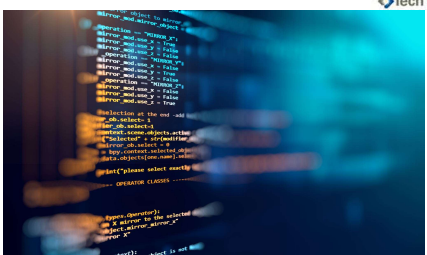
What We Will “Touch On” and Revisit Later

- Deeper recursion in merge sort
- Ranking and similarity
- Hashing and lookup
- Database indexing
- Graph and tree traversal
- Advanced NLP and data-modeling applications



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Review Coding Lab

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Review: What Lab 03 Taught Us

Last week, we focused on:

- How data was stored
- What operation mattered most
- Which structure made the operation clearer
- Which structure made the operation easier or faster

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Today's Question

When is a faster algorithm not actually better?

7

What Counts as Success Today

- Explain the difference between sorting and searching
- Recognize several common sorting algorithms and their basic strategies
- Explain why algorithm choice depends on data size, order, and purpose
- Trace linear search and binary search at an introductory level
- Explain the sorted-data precondition for binary search
- Use evidence to show when an algorithm works, fails, or becomes unreliable

8



Preconditions Make Algorithms Honest

A precondition is something that must be true before the algorithm is valid.

Examples:

- ▶ binary search requires sorted data
- ▶ division requires a non-zero divisor
- ▶ a username lookup requires comparable username values
- ▶ a date sort requires consistent date representation

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Searching And Sorting Work Together

Sorting asks:

- ▶ “How should the data be ordered?”

Searching asks:

- ▶ “How can we find the target?”

The order of the data can change the search strategy.

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Textbook Review

Four Main Areas

The reading covers:

1. sorting algorithms
2. searching algorithms
3. performance analysis
4. practical applications

It also briefly defines Natural Language Processing.

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Textbook Review

NLP Context

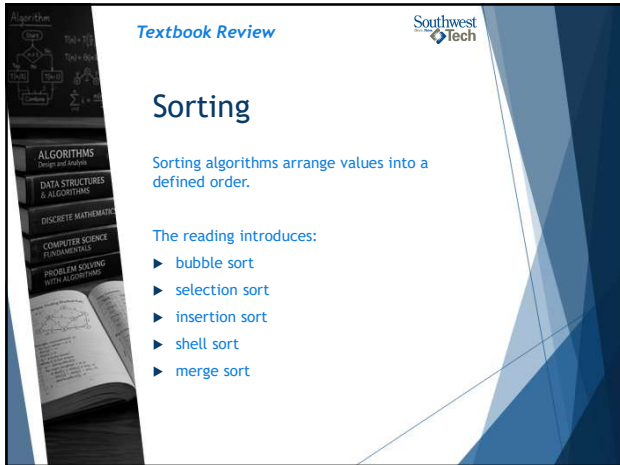
Natural Language Processing is computing work involving human language.

Examples:

- ▶ search suggestions
- ▶ document classification
- ▶ sentiment analysis
- ▶ text ranking
- ▶ chatbot response processing

This week is not an NLP lesson.

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Textbook Review Southwest Tech

Sorting

Sorting algorithms arrange values into a defined order.

The reading introduces:

- ▶ bubble sort
- ▶ selection sort
- ▶ insertion sort
- ▶ shell sort
- ▶ merge sort

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Textbook Review Southwest Tech

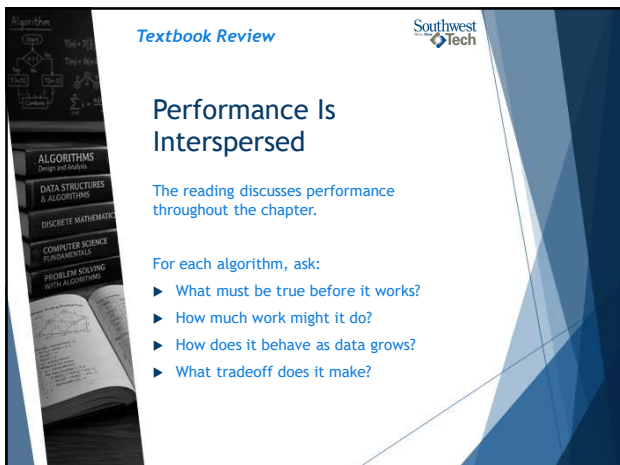
Searching

Searching algorithms try to find a target value.

The reading introduces:

- ▶ linear search
- ▶ binary search
- ▶ interpolation search

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Textbook Review Southwest Tech

Performance Is Interspersed

The reading discusses performance throughout the chapter.

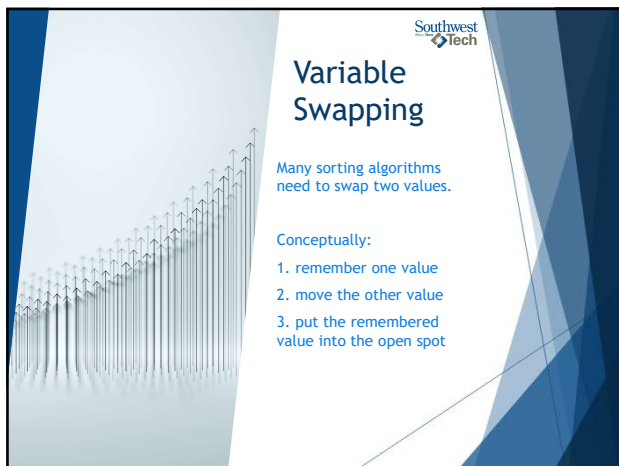
For each algorithm, ask:

- ▶ What must be true before it works?
- ▶ How much work might it do?
- ▶ How does it behave as data grows?
- ▶ What tradeoff does it make?

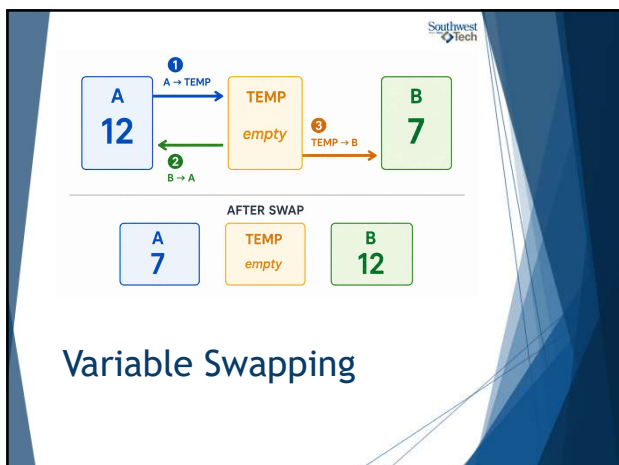
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Bubble Sort

Bubble sort repeatedly compares neighboring values.

If two neighbors are out of order:

- ▶ swap them
- ▶ continue across the list
- ▶ repeat passes until the list is sorted

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Bubble Sort Performance

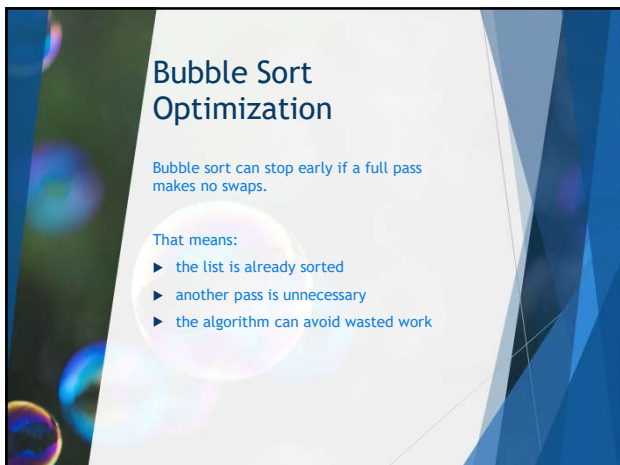
Bubble sort usually performs poorly on larger lists.

Why?

- ▶ it makes repeated passes
- ▶ it compares neighboring values
- ▶ it may keep checking even after little changes

Common growth label: $O(n^2)$

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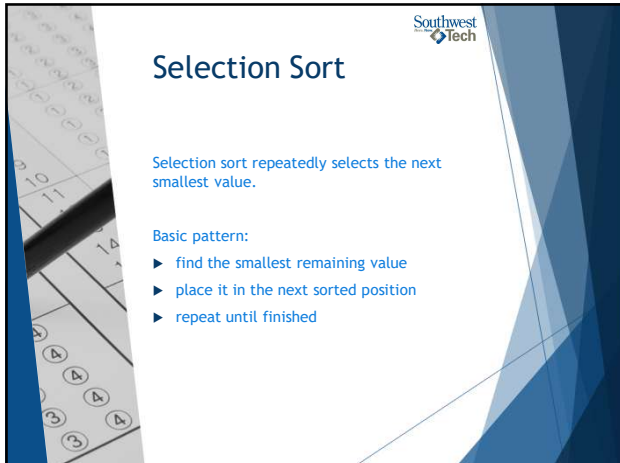
Bubble Sort Optimization

Bubble sort can stop early if a full pass makes no swaps.

That means:

- ▶ the list is already sorted
- ▶ another pass is unnecessary
- ▶ the algorithm can avoid wasted work

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A presentation slide titled "Selection Sort" with the Southwest Tech logo in the top right. The slide features a background image of a calculator. The text explains that selection sort repeatedly selects the next smallest value and lists a basic pattern: find the smallest remaining value, place it in the next sorted position, and repeat until finished.

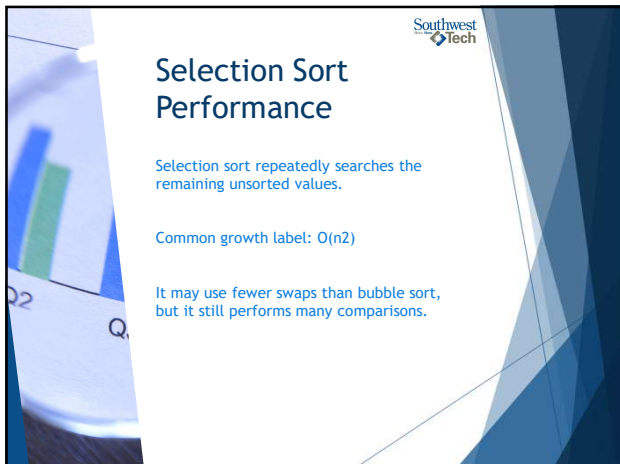
Selection Sort

Selection sort repeatedly selects the next smallest value.

Basic pattern:

- ▶ find the smallest remaining value
- ▶ place it in the next sorted position
- ▶ repeat until finished

22

A presentation slide titled "Selection Sort Performance" with the Southwest Tech logo in the top right. The slide features a background image of a bar chart. The text explains that selection sort repeatedly searches the remaining unsorted values, has a common growth label of $O(n^2)$, and may use fewer swaps than bubble sort but still performs many comparisons.

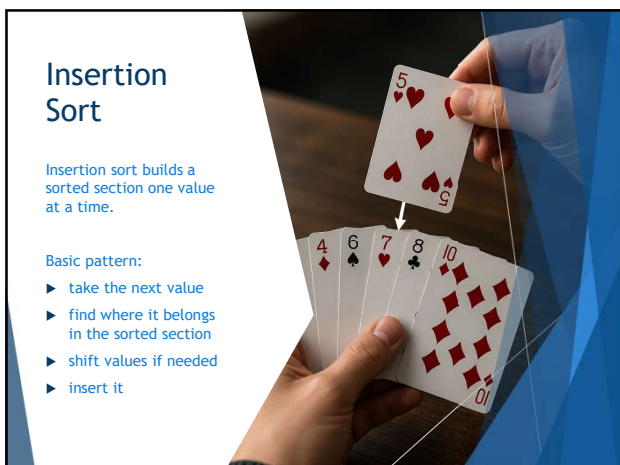
Selection Sort Performance

Selection sort repeatedly searches the remaining unsorted values.

Common growth label: $O(n^2)$

It may use fewer swaps than bubble sort, but it still performs many comparisons.

23

A presentation slide titled "Insertion Sort" with a background image of hands sorting playing cards. The text explains that insertion sort builds a sorted section one value at a time and lists a basic pattern: take the next value, find where it belongs in the sorted section, shift values if needed, and insert it.

Insertion Sort

Insertion sort builds a sorted section one value at a time.

Basic pattern:

- ▶ take the next value
- ▶ find where it belongs in the sorted section
- ▶ shift values if needed
- ▶ insert it

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Insertion Sort Performance

Insertion sort depends heavily on existing order.

It can be:

- ▶ efficient for small lists
- ▶ efficient for nearly sorted data
- ▶ poor for badly unordered larger lists

Common worst-case growth label: $O(n^2)$

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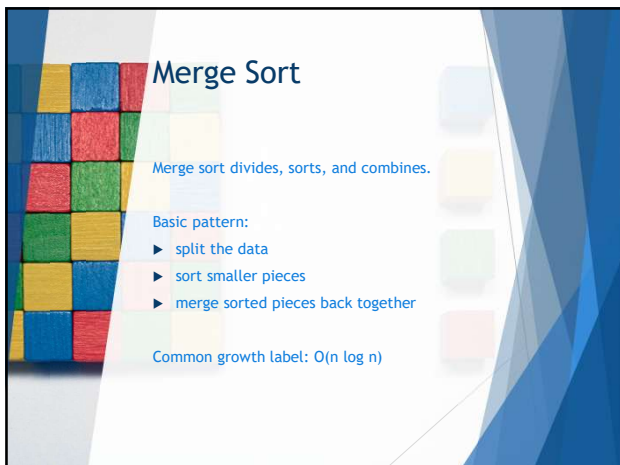
Shell Sort

Shell sort compares values that are farther apart first. Then it gradually reduces the gap.

Purpose:

- ▶ move values closer to their final position sooner
- ▶ finish with smaller local adjustments

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Merge Sort

Merge sort divides, sorts, and combines.

Basic pattern:

- ▶ split the data
- ▶ sort smaller pieces
- ▶ merge sorted pieces back together

Common growth label: $O(n \log n)$

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Sorting Strategy Comparison

Algorithm	Strategy	Taken Away
Bubble	compare neighbors	visible but inefficient
Selection	choose next smallest	organized repeated scanning
Insertion	insert into sorted section	useful for small or nearly sorted data
Shell	compare by gaps	refined insertion idea
Merge	divide and combine	stronger growth behavior

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Sorting Comparison

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Sorting Performance and Choice

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Choosing A Sorting Algorithm

Choosing a sort depends on:

- ▶ data size
- ▶ current order
- ▶ memory limits
- ▶ need for simplicity
- ▶ need for predictable performance
- ▶ whether built-in sorting is acceptable

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Simple Does Not Always Mean Wrong



A simple algorithm can be appropriate when:

- ▶ the data is small
- ▶ clarity matters most
- ▶ the cost is acceptable
- ▶ the goal is learning or verification

But simple should still be explained honestly.

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Performance Claims: Needs Conditions



Avoid

- ▶ "This is faster."
- ▶ "This is better."
- ▶ "This always works."



Use

- ▶ "This is faster when..."
- ▶ "This is valid if..."
- ▶ "This performs better for..."

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Sorting As Preparation

Sorting may be useful before searching when:

- ▶ many searches will happen
- ▶ the data will be reused
- ▶ the search strategy depends on order
- ▶ the sorting cost is worth the later benefit

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Unsorted List

12 17 93 8 55 21

42 → 17 → 93 → ... → 21

Search Item by Item.

Sorted List

8 17 21 42 55 93

8 → 21 → 42 → 55 → 93

Use order to narrow the search.

Sorting can help when lookup happens repeatedly.

Sorting As Preparation

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From Sort To Search

Sorting changes what search can assume.

Unsorted data:

- ▶ Linear search is safe
- ▶ Binary search is not trustworthy

Sorted data:

- ▶ Linear search still works
- ▶ Binary search can use the order

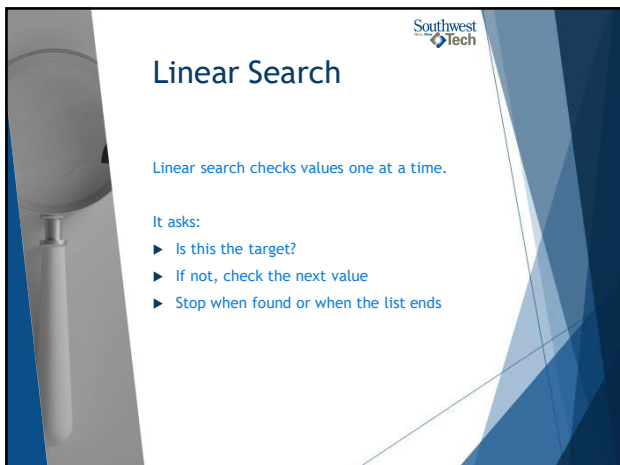
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Linear Search Performance

Linear search:

- ▶ Best case: target is first
- ▶ Worst case: target is last or missing
- ▶ Common growth label: $O(n)$

It may check every value.

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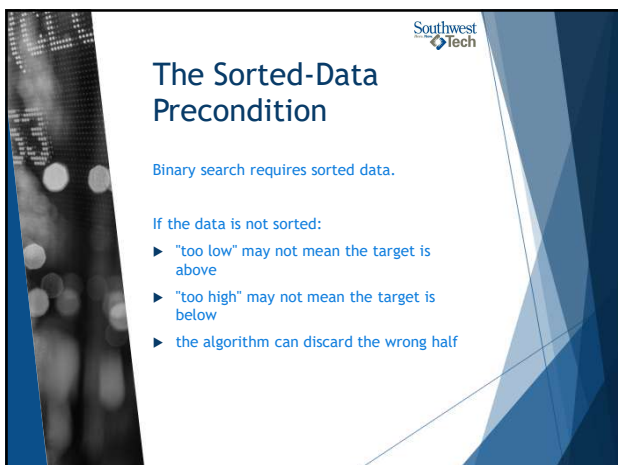
Binary Search

Binary search repeatedly checks the middle.

If the middle value is:

- ▶ the target: stop
- ▶ too low: search the upper half
- ▶ too high: search the lower half

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The Sorted-Data Precondition

Binary search requires sorted data.

If the data is not sorted:

- ▶ "too low" may not mean the target is above
- ▶ "too high" may not mean the target is below
- ▶ the algorithm can discard the wrong half

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Target: 23

LOW MID HIGH

2 5 8 12 16 23 31 44 57

Remaining search range

➔ 23 is greater than 16, so search the right half.

Binary Search

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Step	Low	High	Mid Pos	Value	Decision
1	2	57	5th	16	Search greater half
2	16	57	7 th (3 rd)	31	Search lower half
3	16	31	8th (2 nd)	23	Value found - STOP

Binary Search Trace (Table)

Target: 23

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Interpolation Search

Interpolation search estimates where the target may be.

It works best when:

- ▶ data is sorted
- ▶ values are evenly distributed
- ▶ the target position can be estimated

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Search Strategy Comparison

Search	Needs Sorted Data?	Basic Idea	Common Growth
Linear	NO	Check one at a time	$O(n)$
Binary	YES	Cut search space in half	$O(\log n)$
Interpolation	YES, plus distribution assumptions	Estimate likely position	varies

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Evidence Before Claims

For search algorithms, evidence can include:

- ▶ Test cases
- ▶ Found and Not-found results
- ▶ Trace tables
- ▶ Sorted and unsorted comparisons
- ▶ Explanation of preconditions

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Useful AI Prompt Pattern

"Review this linear search and binary search code. Identify any preconditions. Check for off-by-one risks. Explain whether each test case proves the behavior I claim. Do not rewrite the code unless I ask."

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Records And Keys

Real systems often search records by keys.

Examples:

- ▶ product ID
- ▶ student ID
- ▶ ticket number
- ▶ course code
- ▶ username

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Primary And Foreign Keys

Primary key:

- ▶ uniquely identifies one record

Foreign key:

- ▶ connects a record to another related record

Search and sort help systems find and connect records.

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StudentID	StudentName
101	Maya
102	Jordan
103	Luis

ScoreID	StudentID	LabScore
1	101	92
2	102	85
3	101	96

Matching ID connects the records

Primary Key

Primary Key (unique identifier)

Foreign Key

Foreign Key (references primary key)

Primary and Foreign Keys

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Historical Database Example

The reading's historical database example asks:

- ▶ What identifies a record?
- ▶ What connects records?
- ▶ What should be searched?
- ▶ What should be sorted?
- ▶ What lookup pattern matters most?

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NLP Connection, Briefly

NLP (Natural Language Processing) systems may use search and sort ideas to:

- ▶ find matching terms
- ▶ rank documents
- ▶ sort likely responses
- ▶ retrieve related records
- ▶ compare text features

This is context, not today's coding target.

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Demo Scenario

Problem: Searching book titles on a shelf

Goal Binary Search results on: Sorted vs Unsorted data

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Demo Evidence

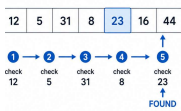
The demo produces:

- ▶ a linear search trace
- ▶ a binary search trace on sorted data
- ▶ a binary search trace on unsorted data
- ▶ a sorted-vs-unsorted summary

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Linear Search Trace

Target: 23



Binary Search Trace

Target: 23



Trace evidence shows how the algorithm reached the answer.

Demo Evidence

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Demo Key Question

Ask after each search:

- ▶ What did the algorithm check?
- ▶ What did it assume?
- ▶ What did it ignore?
- ▶ Was that decision valid?

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Demo Explanation

Example explanation:

"Binary search is valid on the sorted list because each comparison tells us which half can be ignored. On the unsorted list, that decision is unreliable because the values are not arranged in order."

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From Demo to Lab

In Lab 04, you will use a different data set.

You must show:

- ▶ linear search
- ▶ binary search
- ▶ found and not-found tests
- ▶ binary search attempted on unsorted data
- ▶ trace evidence
- ▶ precondition explanation

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Walkthrough Support

The walkthrough artifact can help you:

- ▶ ?

If you use it, you must explain the pattern in your own words, what you changes, and your decisions:

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README Evidence



Your README should show:

- ▶ test case table
- ▶ trace table
- ▶ sorted-data precondition explanation
- ▶ comparison of linear and binary search
- ▶ AI-use note, if used note, if applicable

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AI Use In Lab 04



Manual first:

- ▶ Write or trace your own attempt
- ▶ Create your test cases
- ▶ Identify the precondition

AI-assisted after:

- ▶ Review logic
- ▶ Check edge cases
- ▶ Explain a confusing trace
- ▶ Suggest tests

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Lab On Success Check

- ▶ Searches correctly
- ▶ Includes required test cases
- ▶ Traces at least one search
- ▶ Explains sorted-data requirements
- ▶ Avoids claiming binary search is always better

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What To Carry Forward

Search and sort algorithms teach a larger lesson:

- ▶ Algorithms depend on data conditions
- ▶ Performance depends on context
- ▶ Correctness includes assumptions
- ▶ Evidence makes reasoning visible

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Next Step

1

For Lab 04:

- choose a data set
- implement or simulate linear search
- implement or simulate binary search
- create required tests
- build a trace table
- explain the sorted-data precondition

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How To Use The Textbook For Next Week's Reading



For next week, read for strategy choice.

Focus on:

- ▶ Correctness, performance, and scalability
- ▶ Functional and non-functional requirements
- ▶ Divide and conquer
- ▶ Dynamic programming
- ▶ Greedy strategies
- ▶ PageRank and linear programming as examples

Skim the P, NP-Hard, and NP-Complete section for recognition only.

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Thank you!

Have Fun!

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